



Car Diagnosis Expert System

Using Prolog



Artificial Intelligence - CCE414

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Introduction

The Problem: Car owners often face sudden malfunctions without the expertise to identify the cause, leading to safety hazards and high diagnostic costs.

The Objective: To develop a rule-based expert system that emulates an auto-mechanic's reasoning to troubleshoot common vehicle issues.

Target Users: Everyday drivers and novice mechanics.

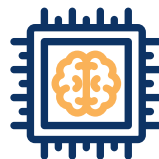
System Architecture

Development Tool:



Built using
SWI-Prolog.

Core Logic:



Utilizes **Backward Chaining**
and **Backtracking** to
traverse the knowledge base.

Optimization:



Implement the (!) Operator to
prevent redundant searches
once diagnosis found.

Diagnostic Rules

1. **Battery:** No Start + Dim Lights
2. **Spark Plugs:** Misfires + High Consumption
3. **Radiator:** Overheating + Coolant Leak
4. **Brake Pads:** Squealing + Poor Braking
5. **Flat Tire:** Flapping + Pulls Sideways
6. **Fuel Pump:** Stalls + Whining
7. **O2 Sensor:** Engine Light + Black Smoke
8. **Alternator:** Battery Light + Electrical Issues
9. **Transmission:** Hard Shift + Red Fluid
10. **Wipers:** Poor Visibility + Streaks



System Workflow (Algorithm)

1. **Initialize:** The user starts the program via the start predicate.
2. **Inquiry:** The system prompts the user with specific symptoms (y/n questions).
3. **Inference:** If symptoms match a rule, the engine identifies the fault.
4. **Result:** The system outputs the Diagnosis and provides Actionable Advice.
5. **Fallback:** If no match is found, a "cannot identify" message is triggered.

Conclusion & Advice Examples

1. **Diagnosis:** Battery Issue → **Advice:** Replace the battery.
2. **Diagnosis:** Radiator Issue → **Advice:** Check hoses and refill coolant.

```
6 ?- start.  
*** Car Diagnosis System ***  
Answer the questions with y. or n.  
-----  
Does the car have (engine_wont_start)? (y/n): y.  
Does the car have (dim_lights)? (y/n): |: y.  
>> Diagnosis: The problem is battery  
>> Advice: Replace the battery.  
true.
```